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Revertible wideband hydrogel-based meta-structure absorber

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Abstract
Multiple bending can have detrimental effect on the performance of the microwave absorber as it may cause fractures and fatigues in the materials it is composed off. Consistency in absorption performance requires performance revertibility along with flexibility. Hydrogels with semi-liquid behavior are found to exhibit reversibility. Herein, meta-structure-based revertible absorbers have been proposed using PVA–borate cross-linked hydrogel popularly known as slime embedded in silicone rubber. Resonators are shaped into cuboidal and hexagonal–prism structures. Dual resonating structure technique has been employed to achieve a wide – 10 dB absorption bandwidth almost covering the entire X-band. The hydrogel with dielectric properties close to water assists in more than 90% absorption over the whole operating band. Simulation shows that the structures are bendable up to a bending radius of 10 mm without compromising on the absorption bandwidth. The self-healing property of slime further aids revertibility of the developed MSAs. Performance restoration is observed after 24 h of continuous application of deformation force for maximum bending. The absorption mechanism in the MSAs are discussed by vector field distribution. Polarization insensitive absorption is found over ±90°.

Keywords Revertible · Hydrogel · Slime · Meta-structure

1 Introduction

Flexible and wearable technology has led to wireless systems which have to go through multiple folding and crumpling during operation. Absorbing shields incorporated with these devices also have to undergo multitude movements and bending, alongwith. Employed shields, while should be conformal, stretchable, have good retracting ability and fast recovery for reliable performance. Flexible electronics are increasingly using hydrogels for its flexibility, compressibility/stretchability and adhesiveness properties [1–5]. Hydrogels are three-dimensional (3D) networks of hydrophilic polymers that can hold a large quantity of water while retaining its structure due to physio-chemical cross-linking of individual polymer chains [6–8], Fig. 1. One of the interesting reported properties of hydrogels is that they return to their initial state on removing the deformation force [3].

Meta-structure-based absorbers are a good replacement to volumetric broadband conventional absorbers. These absorbers are engineered to uniquely amalgamate the sub wavelength resonating periodic structured feature of metamaterial absorbers with the material properties of the resonating structures resulting in broad absorption bandwidth [9–14]. The meta-structure-based absorbers, akin to metamaterial absorber [15], can acquire both positive and negative values of permittivity and permeability [16, 17]. Shape and design of the resonating structure and material properties determine the absorption performance, the operational frequency bandwidth and its polarization sensitivity.

This work proposes development of conformal and revertible meta-structure-based absorbers (MSAs). For this hydrogel-based resonating structures are embedded into flexible matrix. An effective way to achieve wideband absorption is by incorporating multi-resonant structures in a unit cell. MSAs with multi-resonant structures can have two possible configurations: they can be either stacked on top of one another [1–4] or integrated

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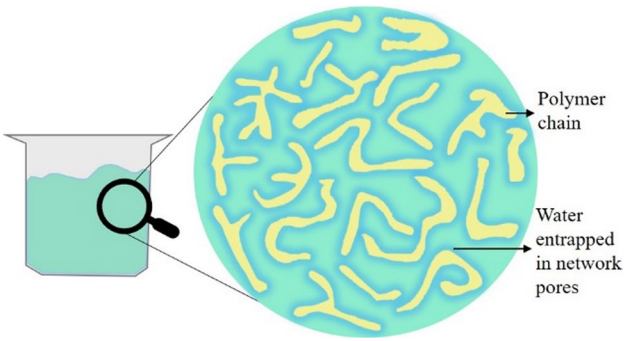


Fig. 1 Schematic of hydrogel

into a co-planar structure [5]. Both the design strategies share the same fundamental element of producing multi-resonance responses that, when coupled, yields a wide-band absorption spectrum. Bandwidth enhancement of the absorber proposed here, is realized by incorporating co-planar embedded resonators at two different frequencies in the same unit structure [18]. The resonators are symmetrically shaped to absorb EM wave at different polarization angles. Prior to fabrication, the dimensions of the unit cell and the resonating structures are optimized to cover the entire working band using CST microwave suite. The fabricated MSAs are tested for absorption performance and evaluated for its extent of restoration ability and recovery time.

2 Hydrogel for embedded resonating structure of meta-structure absorber

Agar and PVA–borate-based hydrogels are initially considered for development of resonating structures. These gels are readily available off-the-shelf and are easy to fabricate. These hydrogels contain ~ 90% of water which makes them lossy at microwave frequencies [13]. Agar-based hydrogel is prepared using 1 wt.% agarose low EEO (from SRL Pvt. Ltd.) in water. The mixture is microwaved for ~ 1 min and allowed to settle for another 10 min. PVA–borate-based hydrogel, commonly called slime, is developed by mixing 5 parts of 4 wt.% PVA (Polyvinyl alcohol) in water solution and 1 part of 4 wt.% borax in water at room temperature. The gel is formed instantly. The synthesized gels are tested for stretchability as well as for their self-healing ability. The gels are poured over petri dish. Both the gels are found to be sufficiently flexible, however, on stretching slime shows better stretchability, Fig. 2a. Self-healing property is tested by slicing the gels into two parts and thereafter placing the edges closely in contact in Fig. 2b. The sliced parts of slime join within 2 s; however, edges of agar-based gel remain separate, as shown in Fig. 2c.

Slime is chosen, here, as material for revertible MSA's resonating structures. The values of complex permittivity (ε) of slime at room temperature, Fig. 3, are measured using a dielectric probe kit (Keysight N1501A) and analyzed using Agilent Vector Network Analyzer E8362C (VNA) in X-band.

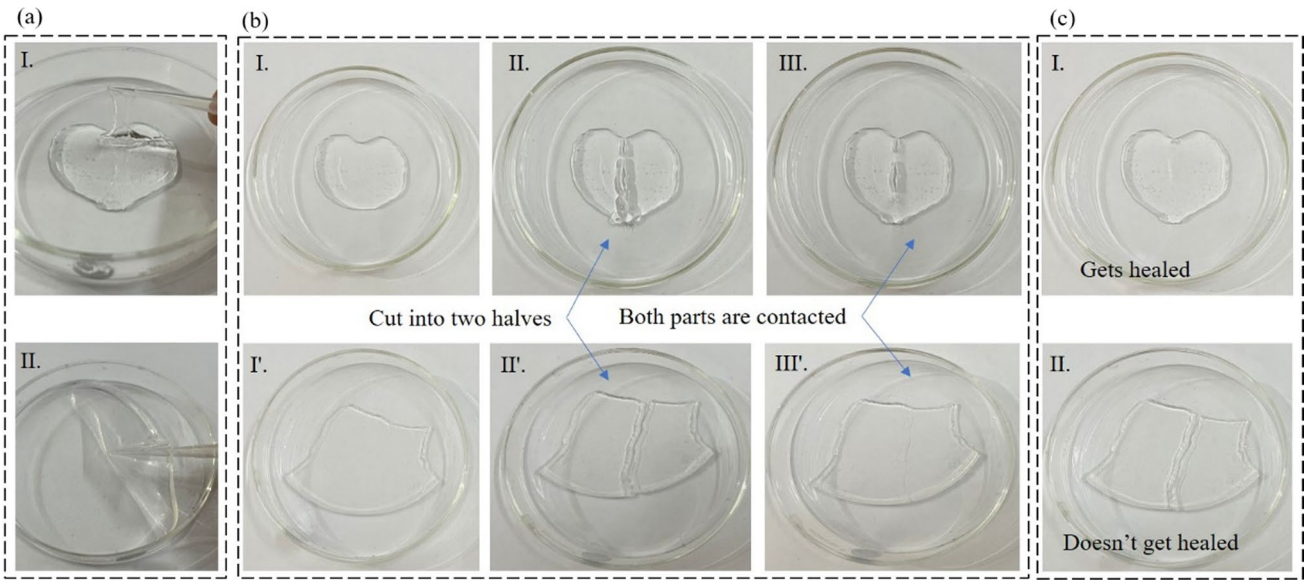


Fig. 2 **a** Stretchability test of I. slime and II. agar-based hydrogel. **b** I., II. and III. slime sliced and brought together and I′., II′. and III′. agar-based hydrogel sliced and edges brought together. **c** I. slime healed after 2 s II. agar-based gel does not heal

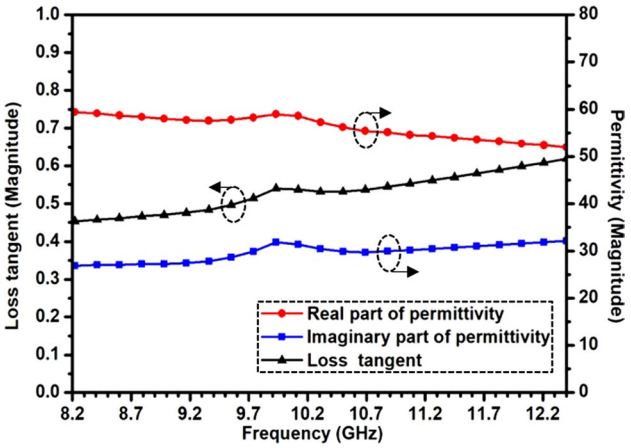


Fig. 3 Complex permittivity values of slime at X-band

A high loss of order of ~0.5 is observed for slime which is close to that of water in the same frequency band [9].

The material characteristics are imported to carry out simulations.

3 Design of wide-bandwidth MSA

Low- and high-frequency resonating structures are selected so as to resonate at either side of the central frequency of the X-band, and thus, the overlapping of the absorbing spectra will cover the entire band. The shapes of the resonating structures of the MSAs are chosen to make it polarization insensitive. Here, two different shaped resonating structures made of slime—cuboidal with fourfold symmetry (4FS) and hexagonal prism with sixfold symmetry (6FS) are used for making the MSAs. Silicone rubber is chosen for embedding the structures. Complex permittivity of silicone rubber is ~ 3.1–j0.12 at the X-band [16]. Hydrophobicity and high temperature resistance of silicone rubber protects the embedded slime resonating structures against external environmental effects [19, 20]. With both slime and silicone rubber being non-rigid the encapsulated MSA design is also flexible. The outer dimensions of the two resonating structures are kept same for easy amalgamation during fabrication. The MSAs are metal backed. Optimizations are conducted to achieve a comprehensive bandwidth with more than 90% absorption, spanning over the X-band. Simulations are carried out employing Floquet port in Frequency Solver Domain on CST Microwave Studio Suite to achieve a wide – 10 dB absorption bandwidth covering entire X-band for both cuboidal MSA 4FS-[L_1L_2] and hexagonal–prism MSA 6FS-[R_1R_2]. Frequency Domain Solver is more useful as it is effective in simulating multi-port systems. In addition, the Solver includes a model-order reduction (MOR) feature which can speed up the simulation of resonant structures such as meta-structures and performance at various incident angles and polarizations for both in TE and TM modes can be extracted directly and simultaneously.

3.1 Geometry of cuboidal MSA

Two different cuboidal shaped resonating structures, [L_1] resonating at higher frequency and [L_2] resonating at lower frequency are combined to form a unit cell, MSA 4FS-[L_1L_2], drawn in Fig. 4a. The parameters, displayed in Fig. 4b, andc, are optimized to cover entire X-band and tabulated in Table 1.

Table 1 Optimized parameters of cuboidal MSA

Parameters	Description	Values (mm)
$l=2a$	Length of the silicone substrate	24
b	Breadth of the silicone substrate	12
a_1	Length of the cuboidal resonator [L_1]	5
a_2	Length of the cuboidal resonator [L_2]	8
$w_1=w_2$	Height of the MSA 4FS-[L_1L_2]	1
g	Silicone layer thickness below and above slime resonators	1
h	Height of the silicone substrate	3
	Copper tape thickness (metal back)	0.035

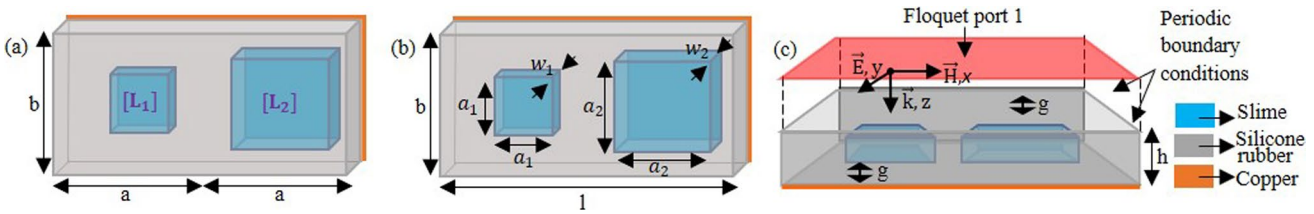


Fig. 4 Schematic of cuboidal MSA 4FS-[L_1L_2] **a** top view layout of unit cell with external dimensions, **b** top view with structural parameters and **c** side view and simulation setup with Floquet port

3.2 Geometry of hexagonal prism MSA

The unit cell here comprises of hexagonal prism-shaped structures, $[R_1]$ and $[R_2]$, resonating at higher and lower frequencies of X-band, respectively. Hexagonal–prism-shaped structure is studied here as the geometry has comparatively smaller total volume with other identical height subwavelength structures resonating at same frequency. The sixfold surface symmetry enables polarization independence over a wider angle. The arrangement and the dimensions of the unit cell, 6FS-$[R_1R_2]$, are given in Fig. 5a–c. The optimized parameters are given in Table 2.

Table 2 Optimized parameters of hexagonal–prism MSA

Parameters	Description	Value (mm)
$L=2A$	Length of the substrate	24
B	Breadth of the substrate	12
r_a	Outer radius of the resonator $[R_1]$	3.4
r_b	Outer radius of the resonator $[R_2]$	5.2
$w_a=w_b$	Height of the MSA 6FS-$[R_1R_2]$	1
G	Silicone layer thickness below and above slime resonators	1
H	Height of the substrate	3
	Copper tape thickness (metal back)	0.035

4 Absorber performance simulation

In simulation, the electromagnetic wave is allowed to incident on the designed absorbers along z-axis (k-vector), electric (E)-field along y-axis and magnetic (H)-field along x-axis, Figs. 4c and 5c using Floquet port. The x- and y-axes of the unit cell are subject to the periodic boundary conditions.

4.1 Formulation of absorption theory

The mathematical expression of absorption $A(\omega)$ [21] is given by

$$A(\omega) = 1 - R(\omega) - T(\omega). \tag{1}$$

The proposed MSAs are metal backed hence the transmittance term, $T(\omega) = |S_{21}|^2 = 0$. Thus, Eq. (1) reduces to

$$A(\omega) = 1 - R(\omega). \tag{2}$$

Now, the absorption will increase on reducing the reflectance term, $R(\omega) = |S_{11}|^2$ alone. Reflectance at normal incidence is given by

$$R(\omega) = \frac{Z_L(\omega) - Z_0(\omega)}{Z_L(\omega) + Z_0(\omega)} \tag{3}$$

where $Z_0(\omega) = 377\Omega$ is the free space impedance and $Z_L(\omega)$ is the absorber impedance and is related to $Z_0(\omega)$ as $Z_L(\omega) = Z_0(\omega)\sqrt{\frac{\mu_{\text{eff}}}{\epsilon_{\text{eff}}}}$. Impedance matching, i.e., $\sqrt{\frac{\mu_{\text{eff}}}{\epsilon_{\text{eff}}}} = 1$, is desired at air absorber interface to minimized the reflectance and this is achieved by tailoring effective permittivity (ϵ_{eff}) and permeability (μ_{eff}) of the absorber.

The normalized impedance at the air–absorber interface can be retrieved from S-parameters using the following equation:

$$Z = Z' + Z'' = \sqrt{\frac{(1+S_{11})^2 - (S_{21}^2 = 0)}{(1-S_{11})^2 - (S_{21}^2 = 0)}} = \sqrt{\frac{(1+S_{11})^2}{(1-S_{11})^2}}. \tag{4}$$

Unity or maximum absorption is achieved when $Z' = 1$ and $Z'' = 0$.

4.2 Broad banding performance

Cuboidal MSA: Figure 6a plots the reflection loss values and corresponding absorbance at normal incidence. It covers a wide − 10 dB bandwidth of 3.8 GHz with $[L_1]$ resonating at 11.43 GHz with reflection loss of − 19.8 dB and $[L_2]$ resonating at 9.03 GHz with reflection loss of − 12.4 dB. Absorbance is found to be > 90% over almost the entire bandwidth, Fig. 6b, using Eq. (2).

Normalized impedance values of the MSAs at X-band are plotted in Fig. 7. The real part, $Z' \rightarrow 1$ while the imaginary

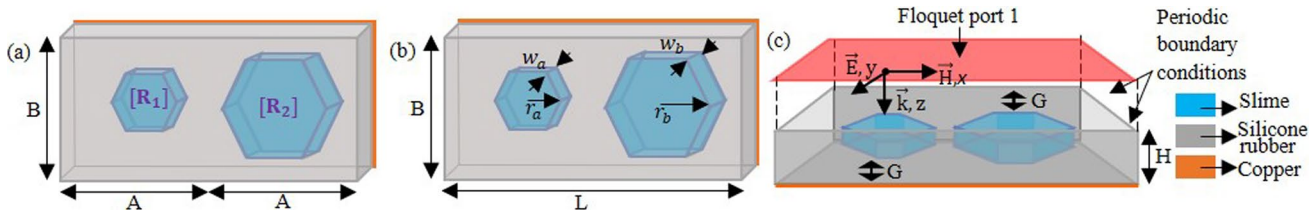


Fig. 5 Schematic of hexagonal–prism MSA 6FS-$[R_1R_2]$ **a** top view layout of unit cell with external dimensions, **b** top view with structural parameters and **c** side view and simulation setup with Floquet port

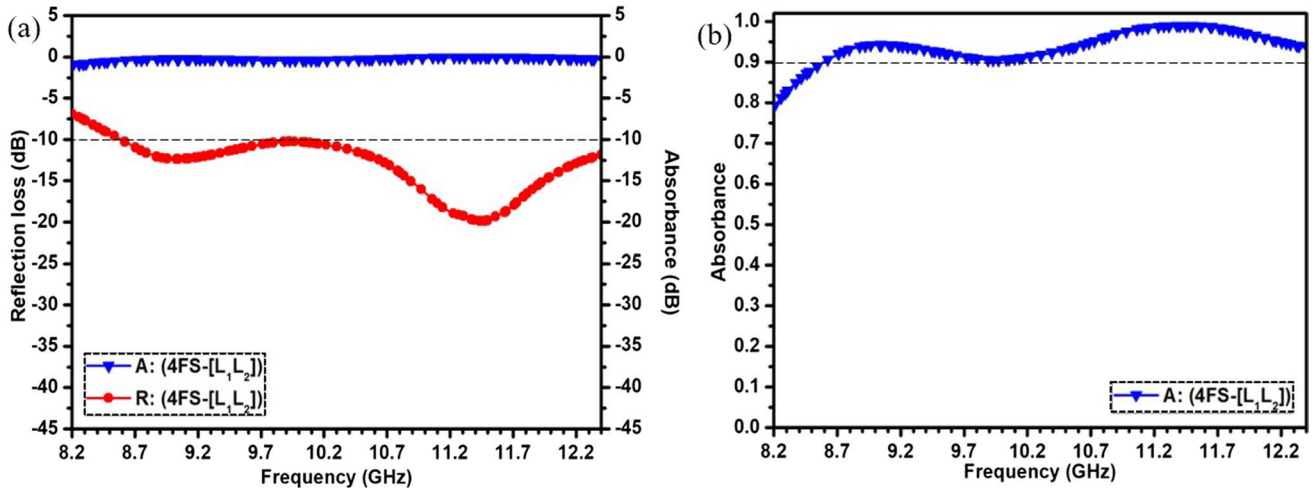


Fig. 6 **a** Reflection loss values with corresponding absorbance in dB scale and **b** absorbance in linear scale of cuboidal MSA

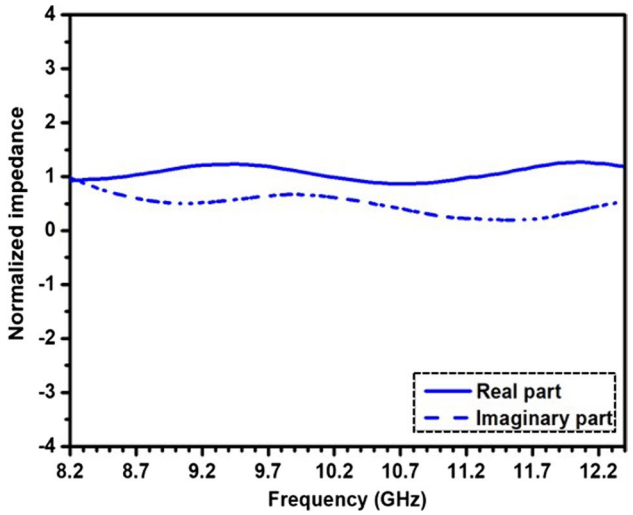


Fig. 7 Simulated normalized impedance of cuboidal MSA

part, $Z'' \rightarrow 0$ almost throughout the band indicating broadband impedance matching.

Hexagonal–prism MSA: The reflection loss and corresponding absorbance spectra at normal incidence are plotted in Fig. 8a. The reflection loss curve shows two resonance peaks at 8.95 GHz and 10.97 GHz with reflection loss of − 12.83 dB and − 17.95 dB, respectively, covering a total – 10 dB bandwidth of 3.7 GHz corresponding to $[R_2]$ and $[R_1]$. Absorbance > 90% is observed over almost the entire bandwidth, Fig. 8b.

Figure 9 displays the normalized impedance values of the MSA at X-band. Broadband impedance matching condition is achieved with Z' approaching 1 and Z'' approaching to zero.

4.3 Field distribution analysis

Comprehensive analysis of vector field distribution is conducted to comprehend the absorption mechanism at resonance peaks.

Cuboidal MSA: Field analysis is carried out at the two dominant resonance frequencies, 11.43 GHz for $[L_1]$ and 9.03 GHz for resonator $[L_2]$. Since the slime's permittivity value is higher than that of the silicone rubber at X-band (Fig. 3), the incident E-field is found to be concentrated towards the edges of the slime resonators, as shown in Fig. 10(a) (i) and (a) (ii). On the other hand, the H-field is localized within the slime resonators, Fig. 10(b) (i) and (b) (ii). As the alternating H-field enters the unit structure, two displacement current loops are generated inside the slime resonators, as shown in Fig. 10(c) (i) and (c) (ii). As can be seen in Fig. 10(d) (i) and (d) (ii), the surface current on the ground plane flows counter to the loop current resulting in magnetic resonance that contributes to the effective permeability which greatly reduces the reflection loss at resonant frequency. Both the induced E-field and the surface current at the metallic plate produces electric response that contributes to the effective permittivity. Figure 10(e) (i) and (e)(ii) shows the scattered power distribution in the MSA indicating the high dielectric loss of the hydrogel contributes to the microwave absorbance.

Hexagonal MSA: Field analysis is carried out for hexagonal–prism MSA, Fig. 11, for resonator $[R_1]$ and $[R_2]$ at frequencies 10.9 GHz and 8.8 GHz. The confinement of the fields shows similar pattern as observed in the previous case with E-field concentrated at the edges and magnetic field confined within the slime resonators, Fig. 11a–d. The lossy slime contributes to the scattered power, Fig. 11e.

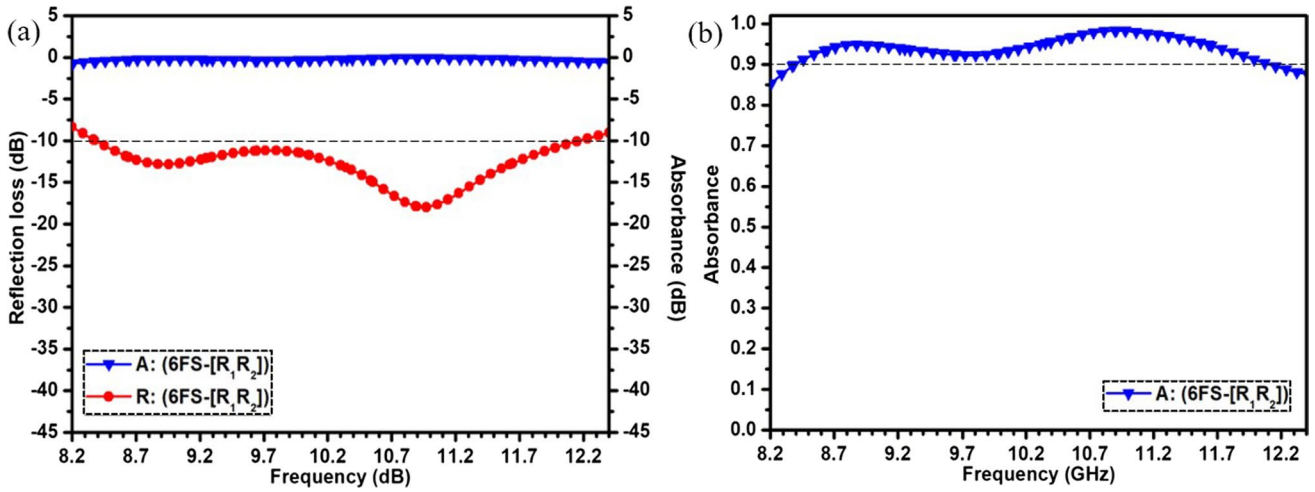


Fig. 8 **a** Reflection loss values with corresponding absorbance in dB scale and **b** absorbance in linear scale of hexagonal–prism MSA

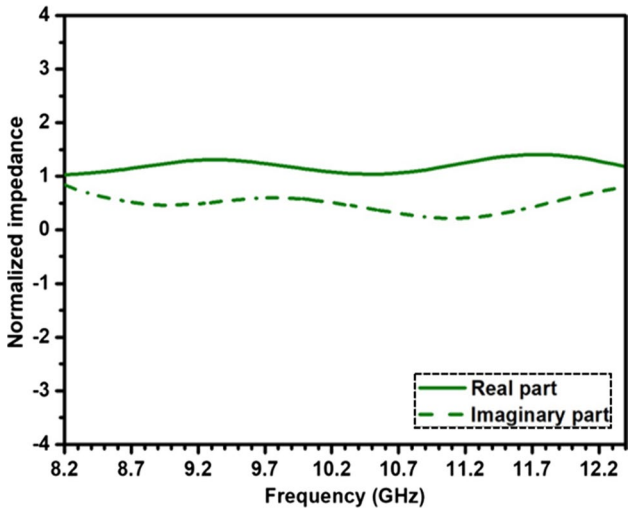


Fig. 9 Simulated normalized impedance of hexagonal–prism MSA

5 Absorption performance-measured

The absorber samples are fabricated using optimized dimensions from Tables 1 and 2. Subsequently, the absorption performance of the developed absorber samples is measured.

5.1 Fabrication

The in-lab fabricated MSAs are sized to $22.86 \times 10.16\ mm^2$ to fit in the waveguide kit. The fabrication is done broadly in three steps—first, moulds for the substrate are crafted using additively manufactured PLA (Polylactic acid), Fig. 12a. The viscous silicone rubber is mixed with 4–5 wt.% of curing agent before pouring into the moulds, and later cured at room temperature for ~ 12 h. In the second step, the synthesized slime is infused inside the grooves as shown in Fig. 12b and left undisturbed for ~ 2 h to settle and take shape. Lastly, for top cover, silicone rubber is poured to fully seal the structure, such that $g/G = 1$ mm. The developed absorber samples are shown in Fig. 12c, d.

5.2 Absorption measurement

The samples are placed in an X-band waveguide kit (Model WR-90X11644A), Fig. 13, that is compatible with the Vector Network Analyzer (Model-E8362) and evaluated using Nicolson–Ross–Weir method [22].

Cuboidal MSA: The MSA is found to show a reflection loss of − 15.47 dB at 8.7 GHz and − 26.4 dB at 11.3 GHz with a − 10 dB bandwidth of 4.2 GHz, Fig. 14a. The entire X-band exhibits an absorption of greater than 90%, Fig. 14b.

Hexagonal–prism MSA: The developed MSA has a reflection loss of − 17 dB at 8.4 GHz and a loss of − 28.9 dB at 10.9 GHz, with a − 10 dB bandwidth of 3.8 GHz, as shown in Fig. 15a. Absorption > 90% is achieved for more than 90% of the X-band as can be observed in Fig. 15b.

5.3 Analysis of the absorption performance

The simulated and experimental absorption results are tabulated in Table 3.

The experimental and simulated results are found to have similar trends covering almost the working band. Variation observed in the results can be attributed to: (1)

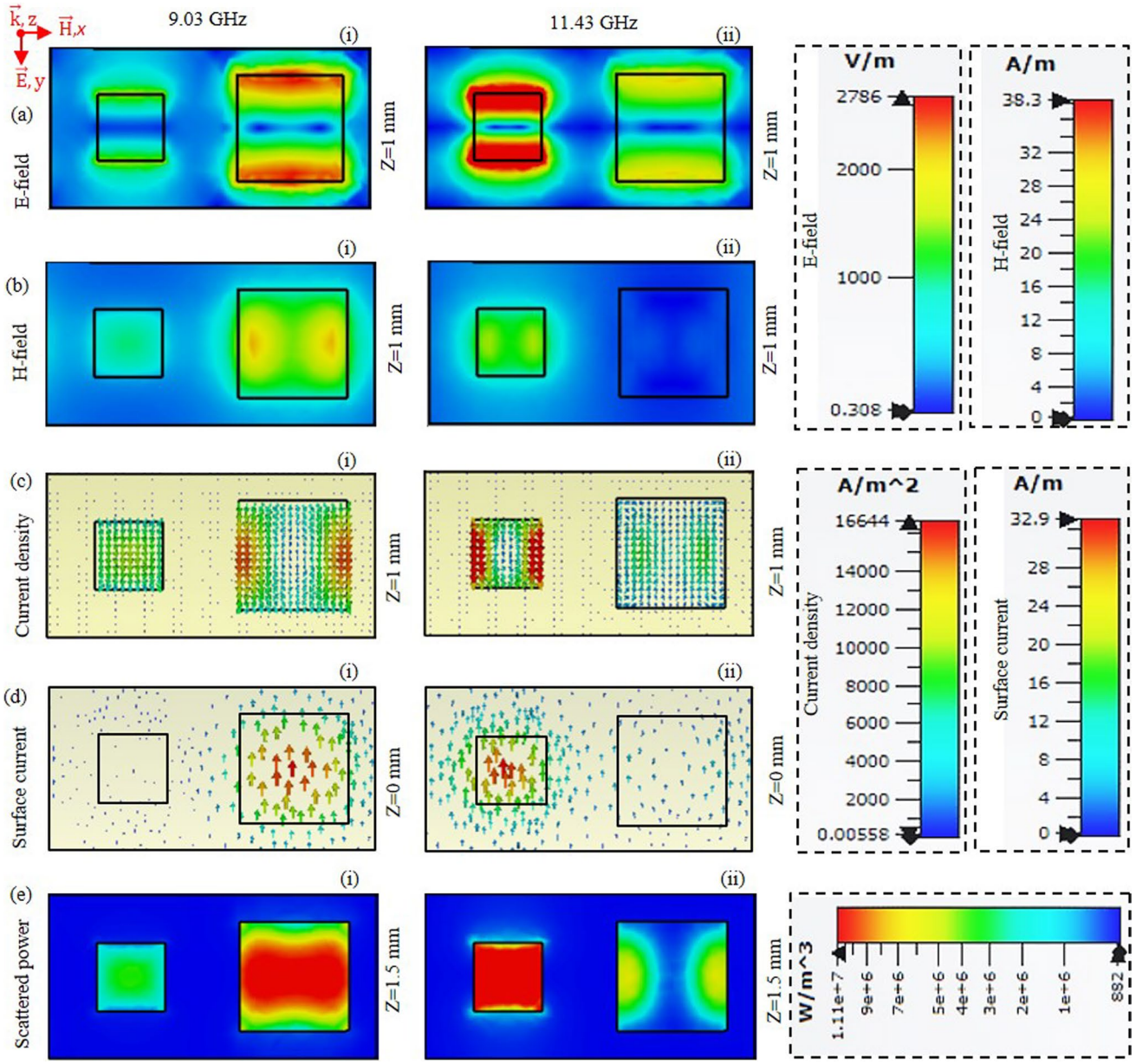


Fig. 10 **a** E-field distribution, **b** H-field distribution, **c** induced current density, **d** surface current on the metallic ground plate and **e** scattered power distribution of cuboidal MSA at frequency 9.03 GHz and 11.43 GHz

van der Waals forces within the slime molecules which are present in realistic slime material; however, there is no provision to take it into account in simulation. (2) In the measurement technique, finite size of the sample, can induce some edge diffractions, which otherwise is taken infinite in simulation. (3) Simulations are carried out using Frequency Solver Domain which is based on finite element method (FEM) that divides a continuous surface into discrete finite elements or meshes and solves polynomial equations at the nodes. A tradeoff between the mesh size and processing time is made to achieve the result which may not necessarily consider microscopic variation in the MSA design that may exists in real, hence the disparity, as observed in Table 3.

6 Revertibility evaluation

A study is conducted on recovery time of the MSAs and the extent of bending it can undergo on application of external deformation force withholding its absorption performance. First, simulations are carried out at different bending

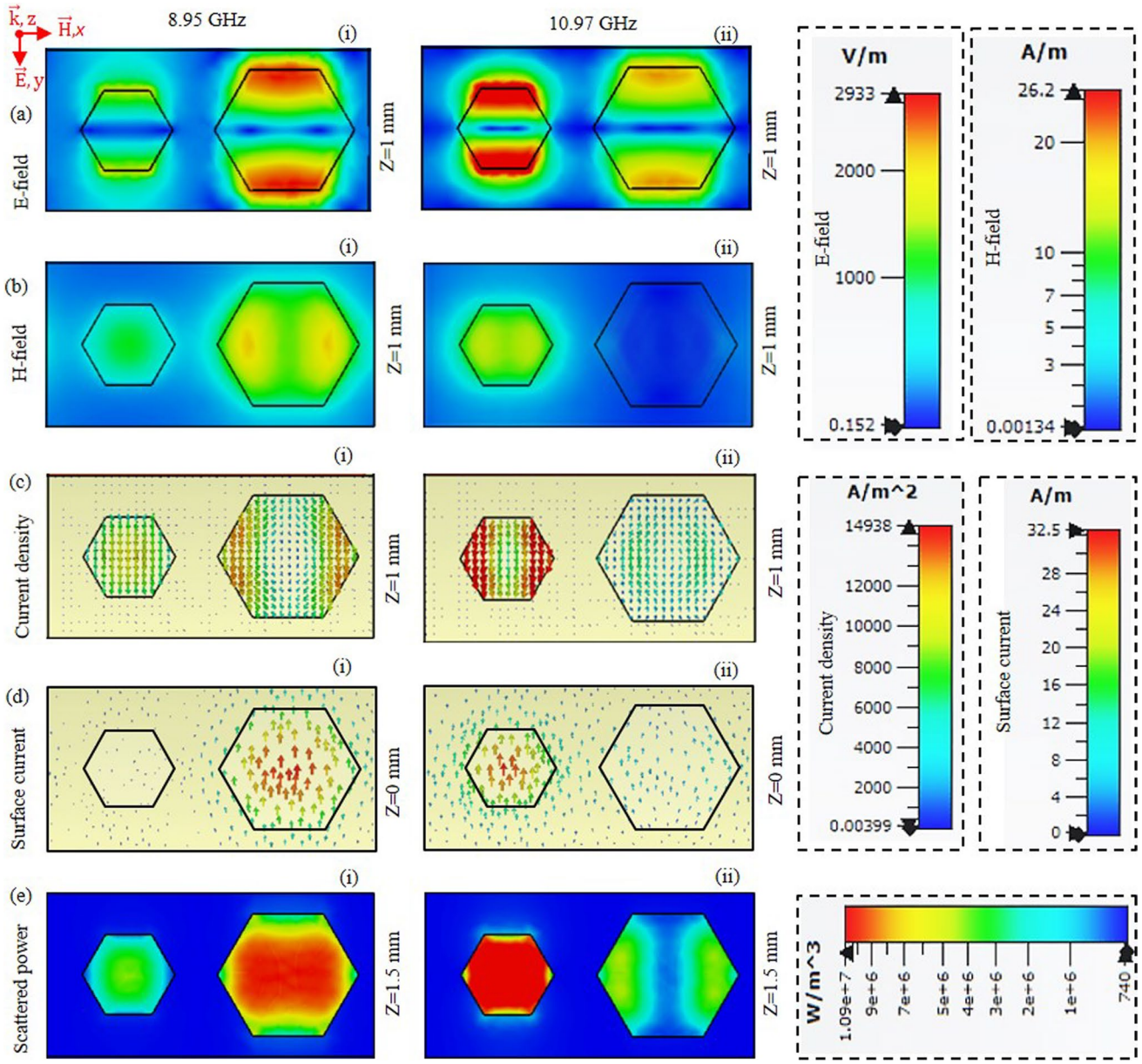


Fig. 11 **a** E-field distribution, **b** H-field distribution, **c** induced current density, **d** surface current at the metallic ground plate and **e** scattered power distribution of hexagonal–prism MSA at frequency 8.95 GHz and 10.97 GHz

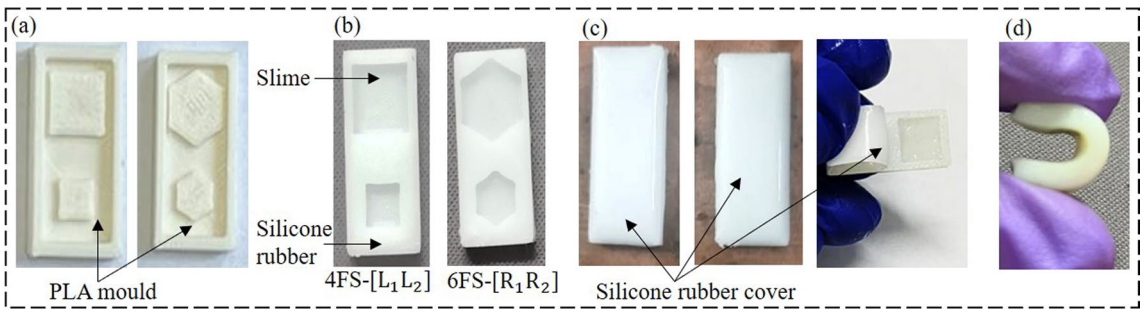


Fig. 12 Fabrication of cuboidal MSA and hexagonal–prism MSA. **a** 3D printed moulds, **b** silicone substrates with grooves filled with slime and **c** sealed final product. **d** Demonstration of flexibility of the fabricated MSA

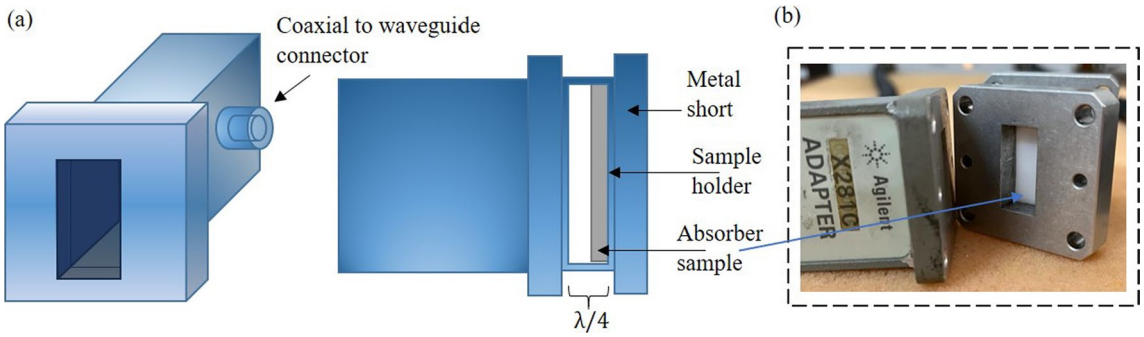


Fig. 13 **a** Schematic of waveguide measurement setup and **b** photograph of MSA mounted inside waveguide adapter

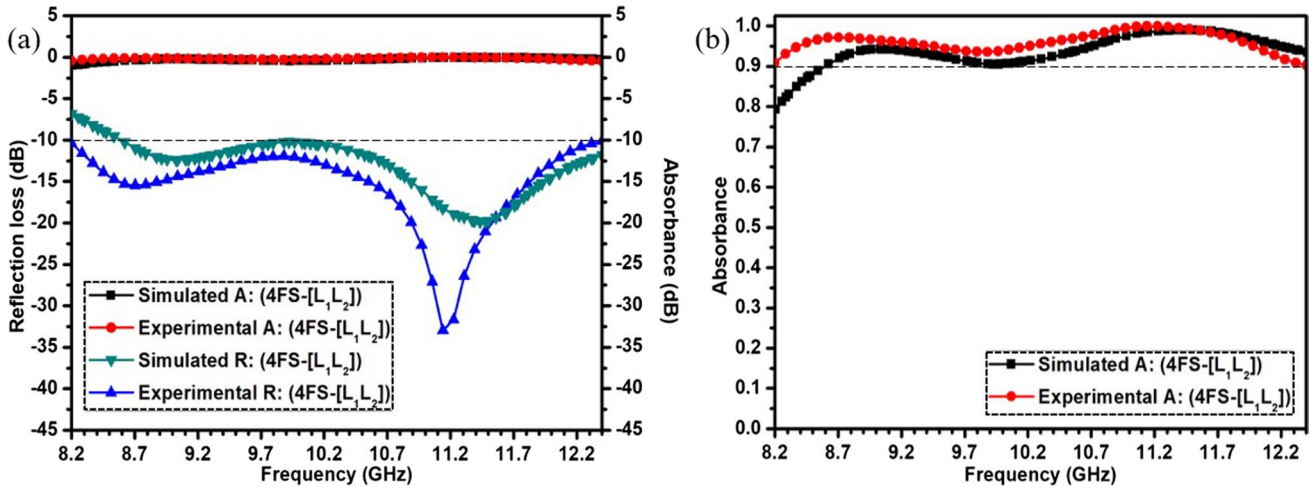


Fig. 14 Measured and simulated values of **a** reflection loss with corresponding absorbance in dB scale and **b** absorbance in linear scale, of cuboidal MSA

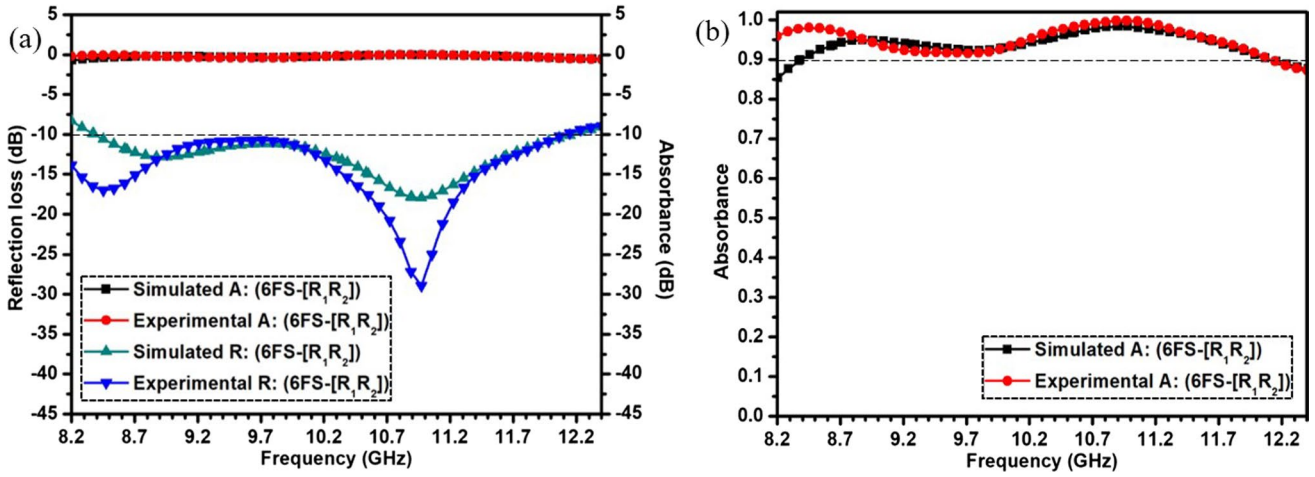


Fig. 15 Measured and simulated values of **a** reflection loss with corresponding absorbance in dB scale and **b** absorbance in linear scale, of hexagonal–prism MSA

Table 3 Absorption results

Absorber	Cuboidal MSA		Hexagonal–prism MSA	
	Simulated	Experimental	Simulated	Experimental
Absorption > 90% bandwidth (GHz)	3.8	4.2	3.7	3.8
% coverage of X-band	90	100	88	90

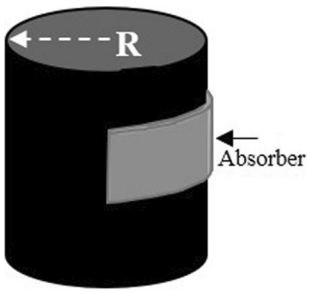


Fig. 16 Schematic of bending radius

curvatures. Figure 16 gives the schematic of the bending curvature, where R is the bending radius. The simulation result gives an idea of maximum bending which can be carried out without effecting the performance of the MSAs. It is observed from the plots in Fig. 17a, b; − 10 dB bandwidth is almost consistent up to $R = 10$ mm. A gradual increase in reflection loss is observed as R decreases from 40 to 10 mm. For $R < 10$ mm, the performance of the absorbers deteriorates drastically. The maximum bending radius limit for the developed absorber prototypes is taken as 10 mm.

A 3D semi-cylindrical support, Fig. 18a for maximum bending, i.e., $R = 10$ mm, is printed using BoXZY 3D printer, as shown in Fig. 18b. The developed MSA is then placed over the cylindrical surface and is secured to the support with the help of thin parafilm M (from Bemis), Fig. 18c. The prototypes are kept in the position for different time durations, thereafter removed and tested instantly for the performance.

It is observed in Fig. 19a, b that the – 10 dB bandwidth tends to take the values close to the simulated values for 10 mm bending radius, Fig. 17a, b; when the absorbers are kept in bending position for 14 h which implies that with increase in the time for bending position, the absorbers take a longer time to recover. After 14 h of bending, no appreciable change in the – 10 dB bandwidth is observed even after 24 h of bending. Hence, studies beyond 24 h is not reported.

The recovery duration of the absorber's performance for $R = 10$ mm after 24 h of bending is observed. The prototypes reverted to its original shape instantly with its – 10 dB absorption bandwidth almost intact. The performance of the absorbers, nevertheless, is restored after 15 min of removal of the deformation force, as shown in Fig. 20.

7 Absorption at different polarization angles

The similar resonators are placed diagonally to develop a symmetrical structured MSA for both cuboidal and hexagonal–prism geometries, as drawn in Fig. 21a, andb. Extent

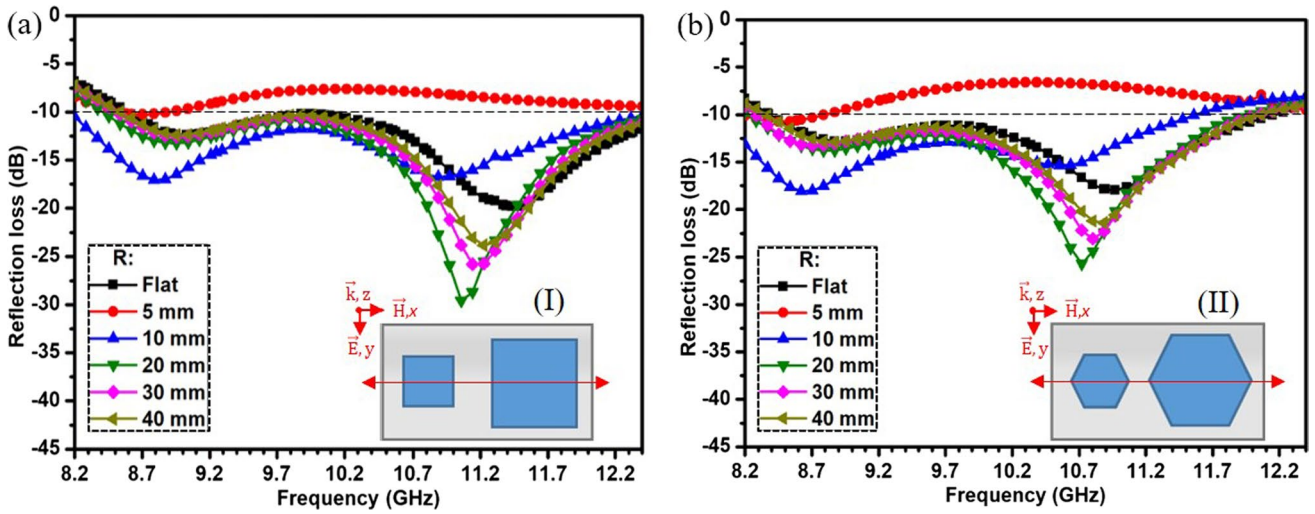


Fig. 17 Simulated reflection loss values at different bending radii for **a** cuboidal MSA and **b** hexagonal–prism MSA. Inset (I) and (II) showing the bending axis

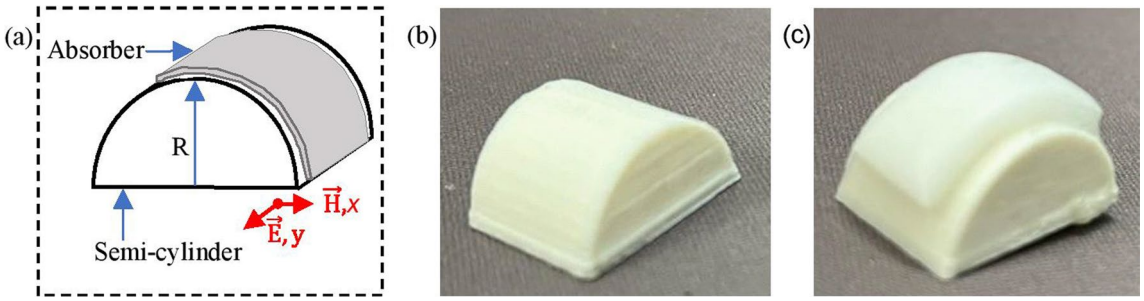


Fig. 18 Semi-cylindrical structure for $R = 10$ mm **a** schematic and **b** 3D printed support **c** with absorber mounted on top of it

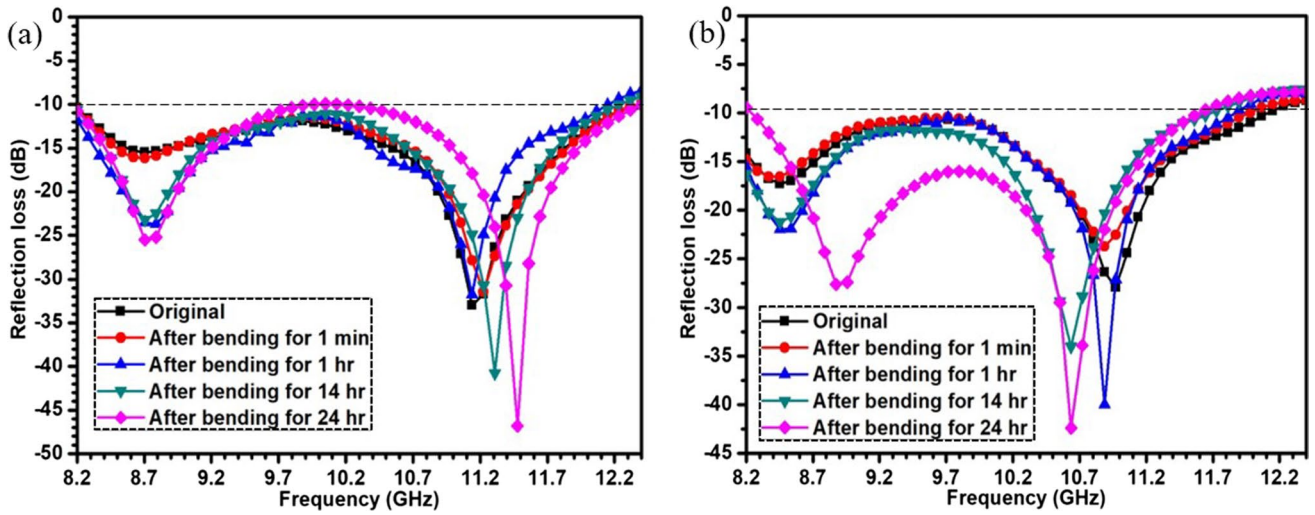


Fig. 19 Measured reflection values of **a** cuboidal MSA and **b** hexagonal–prism MSA after bending for different time duration

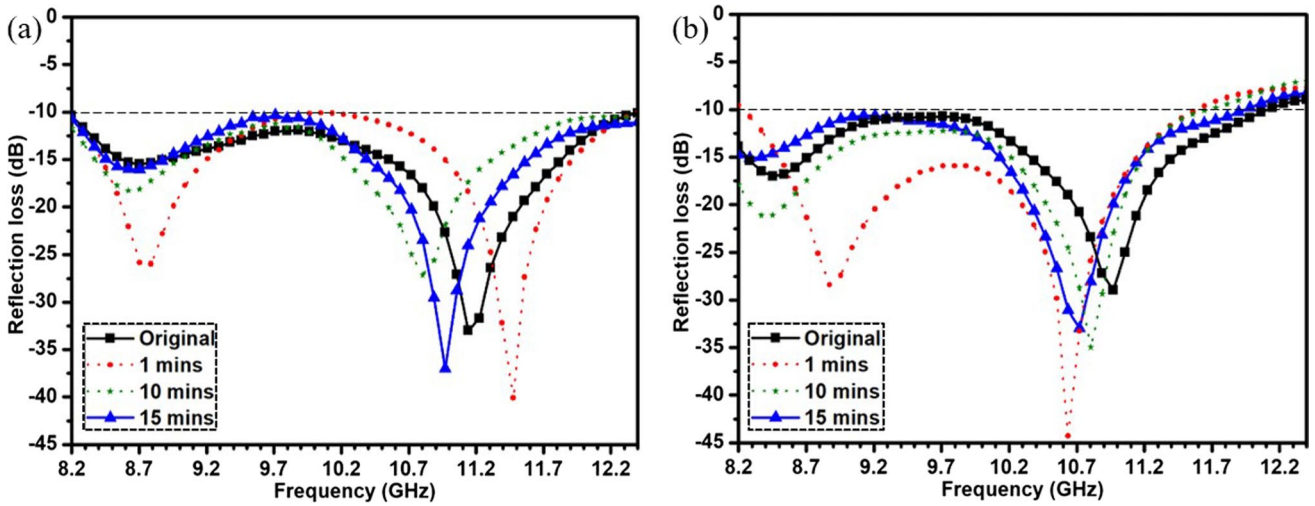


Fig. 20 Recovery of absorption performance in **a** cuboidal MSA and **b** hexagonal–prism MSA

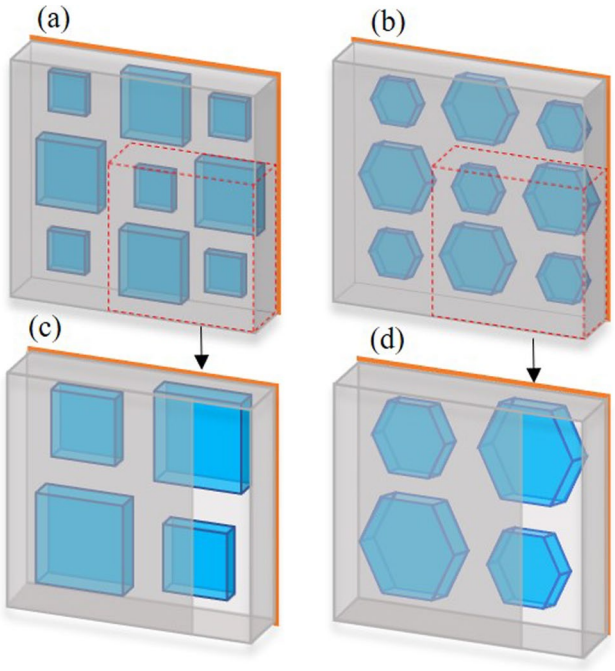


Fig. 21 **a** Proposed cuboidal MSA and **b** hexagonal–prism MSA. Expanded single unit. Expanded single unit cell of **c** cuboidal MSA and **d** hexagonal–prism MSA

of polarization tolerance is studied considering single unit described in Fig. 21c, andd.

The reflection loss values for different polarization angles (φ) from 0°(co-polarized)–90°(cross-polarized) with a step size of 30 are plotted in Fig. 22. Interestingly, the plots coincided for all the polarization angles revealing polarization independency of the MSAs.

The cuboidal MSA is limited up to ±20° of incident angle (θ) for both TE and TM modes, Fig. 23a, b; while in the hexagonal–prism MSA, the − 10 dB bandwidth decreases beyond ±30° of incident angle (θ) for both TE and TM modes, Fig. 23c, d.

Table 4 lists some of the related research on meta-structure-based absorbers. The absorber proposed here consists of embedded resonators. The embedded structures make the absorber relatively thin and gives a planar finish which have added advantage of easy mounting on curved surfaces unlike stacked resonators in [12, 23]. Moreover, co-planar multi resonant structures are easier to design and optimized as the resonant peaks can be independently manipulated with minimal mutual coupling among the resonators [18]. In addition, co-planar embedded structure offers durability to severe environments and reduces installation and fabrication complexity. Use of slime, instead of reported liquid and solid materials [2, 5–9], as embedded resonant structures, is one of the distinguishing features of the work. Slime offers high dielectric loss at microwave frequencies, flexibility, light weight, and ease of synthesis.

8 Conclusion

A revertible meta-structure absorber is proposed in this work. The resonating structures are made up of self-healing slime, embedded in silicon rubber with good elongation at the break (> 600%) enhancing bendability. The flexible feature of the absorbers makes them easy to conform to non-uniform surfaces. The structures are simple and easy to design and fabricate. Dual resonating structures employed in the MSAs exhibit a broad more than 90% absorption

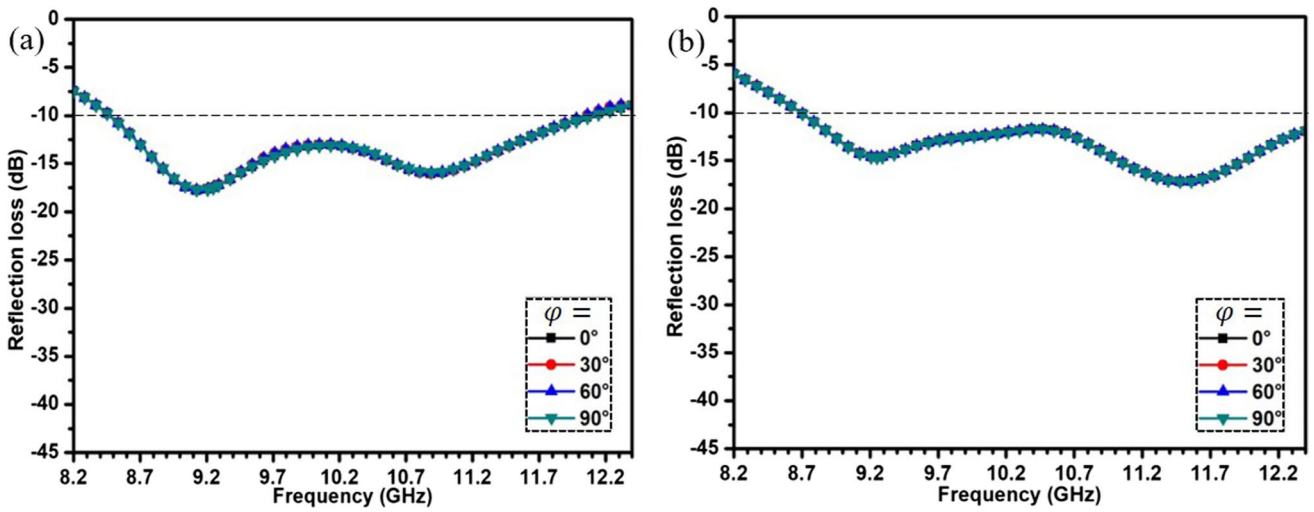


Fig. 22 Simulated reflection loss values at different polarization angles for **a** cuboidal MSA and **b** hexagonal–prism MSA

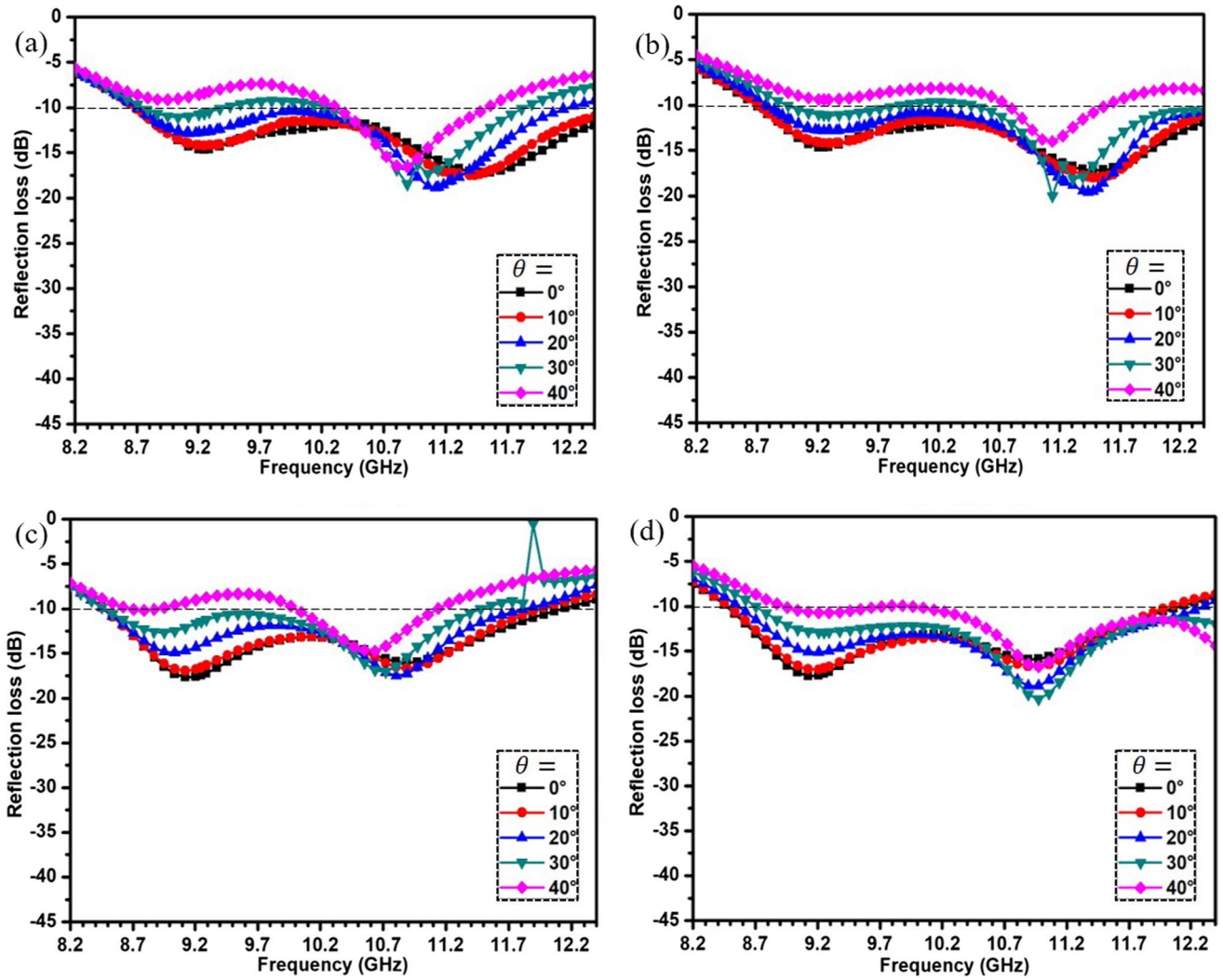


Fig. 23 Simulated reflection loss values at different incident angles for cuboidal MSA **a** TE mode and **b** TM mode; for hexagonal–prism MSA **c** TE mode and **d** TM mode

Table 4 Performance of similar state-of-the-art MSAs

Absorber	− 10 dB operating frequency range (GHz)	Fractional bandwidth (%)	Resonator materials	Flexible/rigid	Revertibility	Thickness mm (λ_0)[a]	Structure design
[18]	8.2–12.4	40.77	Water	Flexible	Not reported	3 mm (0.082λ_0)	Co-planar
[12]	7.6–18	81.25	Graphite SLS	Rigid	–	5 mm (0.13λ_0)	Stacked
[23]	8–18	76.90	Nylon/ carbonyl iron SLS	Rigid	–	2.89 mm (0.077λ_0)	Stacked
[24]	7–40, 75–110	140.4, 37.8	CC/Wax	Rigid	–	11 mm (0.262λ_0)	Stacked
[10]	2–4, 8–40	66, 133	Carbon composite	Flexible	–	5 mm (0.033λ_0)	Stacked
Proposed MSAs	8.2–12.4	40.77	Slime (PVA–borate hydrogel)	Flexible	Revertible	3 mm (0.082λ_0)	Co-planar

[a] λ_0—lowest operating frequency

Table 5 Summary of the absorbers' specifications

Absorber	− 10 dB absorption bandwidth (GHz)	Maximum bending radius (mm)	Recovery time[a] (Min/Max)	Polarization angle tolerance	Incident angle tolerance	Water absorption[b] (%)	Density (g cm^{-3})
Cuboidal MSA	4.2 (8.2–12.4)	10	1–15 min	±90°	±20°	0.6	~1.16
Hexagonal–prism MSA	3.8 (8.2–12)	10	1–15 min	±90°	±30°	0.6	~1.16

[a]For 24 h of bending with radius—$R = 10$ mm

[b]Water absorption= $\frac{\text{Weight}_{\text{submergered}} - \text{Weight}_{\text{actual}}}{\text{Weight}_{\text{actual}}} \times 100$

bandwidth covering almost entire X-band. The co-planar resonators unlike multi-stacked resonators do not add thickness to the absorber. The efficacy of the absorber is consistent even after 24 h of bending with bending radius of 10 mm. The MSAs show a minimum recovery time of ~1 min on 24 h bending with performance getting completely reverted after ~15 min. Fourfold (cuboidal) and sixfold (hexagonal–prism) symmetric resonators enable polarization independent absorption. The absorption performance is insensitive to wide incidence angle. The performance parameters are summarized in Table 5.

The absorbers are resistant to water, relatively thin with thickness ~$\lambda/10$. Light weighted and revertible finding ample applications in flexible and wearable technology. The non-toxic biocompatible nature of the material used can make it viable to be used in proximity to human. Low cost of materials and additive manufacturing molds minimizes the fabrication time making the absorber practically realizable.

Author contributions All authors contributed to the study. Design and fabrication are performed by SS and HS. Data analysis is done by SS and supervised by Dr. NSB. The manuscript is drafted by SS and HS. Dr. NSB provided critical revision of the manuscript. All authors read and approved the final manuscript.

Data availability All data generated or analyzed during this study are included in this article.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethics approval Not applicable.

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